

LAWE-CL2: Multi-agent LLM-based automated writing evaluation system integrating linguistic features with fine-tuning for Chinese L2 writing assessment

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ABSTRACT

The recent emergence of Large language models (LLMs) has opened up new avenues for the advancement of English L2 automated writing evaluation (AWE). However, such research for Chinese L2 AWE remains scarce. Existing LLM-based AWE research neglect the integration of scoring and feedback generation, facing challenges such as limited reasoning ability from linguistic perspectives, potential unfairness arising from single-agent evaluation, and the lack of a trainable pipeline for generating scores and feedback. To bridge the gaps, this study proposes LAW-CL2, an LLM-based approach for Chinese L2 AWE that integrates multidimensional linguistic features into prompts, simulates collaborative scoring through a multi-agent framework, and enhances model performance through synthetic feedback and fine-tuning. Extensive experiments conducted in this study on the HSK Dynamic Composition Corpus show three key findings: (1) the incorporation of detailed information from linguistic features into prompts substantially improves both scoring and feedback performance; (2) the multi-agent framework contributes to alleviating unfairness and inconsistency in both scoring and feedback; (3) fine-tuning on synthetic feedback enables the LLMs to achieve state-of-the-art AWE performance, outperforming mainstream baseline models. Notably, the feedback generated by the models still has instructive value even when the scoring prediction is inaccurate.

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1. Introduction

Artificial intelligence (hereafter AI) is increasingly applied in education (Yang et al., 2025), particularly in automated writing evaluation (AWE) of second language writing (L2 writing) (Shermis & Burstein, 2013). AWE systems provide essay scoring and diagnostic feedback, offering objectivity and timely support for recursive writing (Shermis & Wilson, 2024; Tate et al., 2024). These functions make AWE an important tool in L2 acquisition. The recent emergence of Generative AI (hereafter GenAI) has opened up new avenues for the advancement of L2 AWE. Large language models (hereafter LLMs), such as ChatGPT, demonstrate remarkable abilities in language comprehension and text generation. These advances have ushered L2 writing research, teaching, and learning into a new phase (Yang & Chen, 2025). LLMs can perform essay scoring and feedback generation under zero-shot or few-shot conditions, and their user-friendly interactive capabilities substantially expand the applicability and potential of AWE.

Existing LLM-based L2 AWE research has predominantly focused on English L2 writing, while corresponding studies on Chinese L2 AWE remain at an early stage. This imbalance limits the accumulation of empirical evidence from other languages, making it difficult to substantiate the generalizability of LLMs in writing assessment (Lan et al., 2025). Recently, there has been a steady growth in the population of learners acquiring Chinese either as a second language or as a heritage language⁶ (Yan & Lin, 2023). Chinese has been incorporated into the national education systems of 86 countries worldwide,⁷ and the number of international Chinese learners and users has surpassed 200 million.⁸ The increasing reach of the *Hanyu Shuiping Kaoshi* (Chinese Proficiency Test, hereafter HSK) has been accompanied by a steady rise in the number of candidates internationally (Teng, 2017; Peng et al., 2020). Therefore, given the vast population of Chinese L2 learners, there is an urgent need for efficient, standardized tools for Chinese L2 writing assessment. In the absence of such tools, Chinese L2 teachers face an increasing grading burden. Meanwhile, students lack timely and personalized feedback that captures the distinctive characteristics of Chinese writing, which is crucial to the process of second language acquisition (Wang & Gayed, 2024). Furthermore, the lack of such tools runs counter to the current trend of AI-assisted education. It may limit the quality of Chinese L2 education, make it difficult to meet the growing global demand for Chinese learning, and hinder progress in deepening our understanding of Chinese L2 acquisition.

To address these open issues, this paper introduces LAWE-CL2, a methodological approach to LLM-based automated writing evaluation for Chinese L2 writing, integrating multidimensional linguistic features, multi-agent collaboration, and fine-tuning. LAWE-CL2 can simultaneously provide timely scores and evaluative feedback, with the potential to enhance the integrity of assessment and its educational value. However, three main challenges remain. First, it is essential to investigate how LLMs can be effectively aligned with the scoring standards for Chinese L2 writing and how the relationship between essay scores and linguistic features can be systematically established. Second, most existing studies rely on single-agent scoring and fail to replicate the multi-rater plus arbitration mechanisms that are widely adopted in operational writing assessment contexts, such as TOEFL and IELTS. Third, few studies have fine-tuned LLMs on AWE data for scoring and feedback generation. In particular, fine-tuning that integrates both scoring and feedback tasks remains scarce, thereby constraining the generalizability of the models. These challenges not only reveal the shortcomings of previous research but also motivate the methodological entry points and innovations of the present study.

2. Literature review

2.1. A brief review of the Pre-GenAI era of AWE

AWE involves assigning scores to essays and providing students with diagnostic and formative feedback to help them understand the strengths and weaknesses of their writing (Grimes & Warschauer, 2010). The scoring functions of early AWE systems primarily relied on manually defined linguistic features and rule-based methods, with scores subsequently mapped through traditional statistical models (Page, 1966; Attali & Burstein, 2006). Supervised machine learning has improved predictive performance and modeled nonlinear feature relationships (Mohler & Mihalcea, 2009; Persing & Ng, 2013). Although these approaches relied on manual feature engineering and thus offered strong interpretability (Rudner & Liang, 2002), they also faced clear limitations. In particular, they were unable to capture more complex aspects of writing such as discourse organization and deep semantic features (Lu, 2021), resulting in restricted feature coverage and limited generalizability across languages and corpora. In the past decade, AWE has shifted from traditional machine learning to deep learning, enabling automatic semantic representation learning and reducing reliance on manual feature engineering (Taghipour & Ng, 2016; Sun et al., 2025). According to the reviews by Ramesh and Sanampudi (2022) and Misgna et al. (2024), such progress has been observed across different types of language models, ranging from early architectures represented by RNNs, LSTMs, and CNNs to Transformer-based models such as BERT. Nevertheless, their black-box nature limits interpretability and reduces transparency in scoring decisions.

Evaluation feedback in AWE is primarily used for formative assessment in instructional contexts and can be generated in multiple formats (Burstein et al., 2013; Knight et al., 2020), offering substantial pedagogical utility. It can enhance instructional efficiency and support language learners by helping them address writing problems promptly, foster autonomy, and sustain motivation (Wang et al., 2013; Shi & Aryadoust, 2024; Sari & Han, 2024), thereby improving overall writing quality. Early systems, however, relied on rule-based detection and template-based responses, such as Criterion®, which produced mechanical feedback that lacked the

⁶ In this paper, we refer to them uniformly as Chinese L2 learners for convenience.

⁷ <https://www.chinanews.com.cn/edu/2024/09-26/10292745.shtml> (accessed 27 September 2025)

⁸ <https://m.chinanews.com/wap/detail/cht/zw/10401481.shtml> (accessed 27 September 2025)

diagnostic value provided by human teachers (Burstein et al., 2003). Meanwhile, prior research cautions that such systems often conceptualize writing as the mastery of discrete subskills and place particular emphasis on grammatical accuracy (Hyland & Hyland, 2006). In practice, mainstream tools commonly flag potential problems and offer explanations and correction suggestions, but their feedback often has limited error coverage and can be vague or insufficiently actionable (Shadiev & Feng, 2024; Ding & Zou, 2024). Moreover, the lack of human interaction can limit individualized and communicative feedback (Liao, 2016). Although deep learning has substantially advanced AWE, progress in generating high-quality essay feedback has remained limited for a considerable time. This is because encoder-only language models such as BERT and RoBERTa are primarily designed for text classification and understanding tasks (Minaee et al., 2024), making them unsuitable for comprehensive text generation. Although encoder-decoder models such as T5 and BART possess some text generation capability, their relatively small parameter scales and limited context length pose challenges for producing detailed writing feedback.

2.2. The Potential of Using GenAI in L2 AWE

2.2.1. Automated scoring for L2 writing

Recently, researchers have begun exploring GenAI tools such as ChatGPT in English L2 AWE as a way to address the limitations of approaches developed before the emergence of GenAI. These methods generally rely on designing elaborate prompts that embed detailed scoring rubrics, thereby guiding LLMs to better align with rating criteria, while leveraging zero-shot or few-shot settings to accomplish automated scoring. For example, Mizumoto and Eguchi (2023) pioneered the use of text-davinci-003 for TOEFL11 essay scoring in a zero-shot setting, reporting strong score stability across repeated runs (QWK = 0.682) and suggesting that LLM-based scoring could complement human ratings. Pack et al. (2024) evaluated PaLM 2 (Bard), Claude 2, GPT-3.5, and GPT-4 on ESL placement essays using a zero-shot rubric-in-prompt design, and repeated the ratings at two time points at least 90 days apart. They found that GPT-4 performed best, showing strong test-retest stability (intrarater ICC: T1 = 0.897, T2 = 0.927) and solid agreement and correlation with human ratings (LLM-human ICC: T1 = 0.843, T2 = 0.779; r : T1 = 0.731, T2 = 0.638).

Studies generally find that, compared with zero-shot prompting, few-shot prompts with a small set of pre-scored reference essays help LLMs calibrate to rubric levels and typically improve scoring performance. For example, Yancey et al. (2023) evaluated the performance of GPT-3.5 and GPT-4 on short writing responses from the Duolingo English Test, using a few-shot prompting strategy that combined a minimal rubric with a small set of calibration examples. Their results showed that GPT-4 achieved substantial agreement (QWK = 0.81), approaching the performance of a traditional scoring model (QWK = 0.84). Kim (2025) further reported that the best performance was obtained when detailed scoring rubrics, explicit reasoning requirements, and few-shot exemplars were used in combination (r = 0.61, QWK = 0.45).

An emerging trend is to integrate multidimensional linguistic features into LLM-based AWE. Recent studies have operationalized this approach by incorporating GPT scores and linguistic indices into regression models. Mizumoto and Eguchi (2023) demonstrated overall improvements in predictive performance, while Liu et al. (2025b) found gains primarily for low-proficiency essays, suggesting that such integration can help mitigate class imbalance. Extending this line of research, Li and Liu (2024) embedded linguistic indices into prompts to guide GPT-4, and found that using all indices yielded the best proportional reductive mean square error (PRMSE), outperforming baselines and other models.

However, even with carefully optimized prompting strategies, the scoring performance of non-fine-tuned LLMs remains limited (Liu et al., 2025a). Fine-tuning refers to additional training of LLMs on task- or domain-specific data to enhance performance in the target context (Brown et al., 2020). Recent evidence suggests that fine-tuned LLMs hold promise for AWE (Latif & Zhai, 2024). For instance, Wang and Gayed (2024) fine-tuned GPT-3.5 on the TOEFL writing corpus and found that, compared with zero-shot prompting, fine-tuning enhanced the performance of LLMs in automated scoring of L2 essays (QWK = 0.78). Johnsi and Kumar (2025) fine-tuned Llama-3 on a subset of the ASAP dataset and achieved a mean QWK of 0.72 on the test set. The fine-tuned model outperformed a baseline LSTM with tuned hyperparameters and Word2Vec embeddings (QWK = 0.596), suggesting that task-specific fine-tuning can yield stronger AES performance than prompt-only approaches. Liu et al. (2025a), (2025b) reported similar findings, suggesting that fine-tuning is likely to play an increasingly important role in future AWE research. For example, Liu et al. (2025b) reported that the fine-tuned GPT model achieved an overall exact agreement rate of 78.3% with human raters.

2.2.2. Automated Feedback for L2 Writing

Although LLM-generated scores remain difficult to interpret, they can serve as complementary tools for formative and low-stakes assessment, since their feedback remains educationally valuable (Yeung, 2025). Barrot (2023) preliminarily explored the instructional role of ChatGPT in L2 writing, highlighting its potential for formative feedback. Subsequent empirical studies have largely followed two lines of research. One strand evaluates the validity of LLM-generated feedback by benchmarking it against human coding, teacher feedback, or rubric-based criteria, typically without involving learners (Steiss et al., 2024). The other strand examines how learners receive, interpret, and use LLM-based automated feedback during drafting and revision in L2 English writing contexts (Abdi Tabari et al., 2025). Given our research focus, we primarily review studies in the first strand. Previous empirical research has pursued three related directions. First, a substantial body of research has examined whether LLMs can reliably diagnose and correct writing errors by benchmarking model outputs against human coding. For example, Pfau et al. (2023) reported a strong correlation between ChatGPT's error detection and human coding ($\rho > 0.90$). Mizumoto et al. (2024) reported that ChatGPT performed comparably to human evaluators in error detection, showing a stronger association with human ratings ($\rho = 0.79$) than Grammarly ($\rho = 0.69$). Second, studies have examined LLM feedback quality. Guo and Wang (2024) compared feedback from ChatGPT and EFL teachers on argumentative essays written by Chinese undergraduates, focusing on content, organization, and language. The results showed that ChatGPT

produced more comments and a more balanced distribution across dimensions, whereas teacher feedback concentrated mainly on content and language. Wang et al. (2025) found that LLMs can generate multidimensional feedback grounded in assessment criteria. Gjorevski et al. (2025) examined the dual potential of ChatGPT for scoring and feedback in English essay assessment. They found that ChatGPT showed relatively high agreement with human raters in score assignment. It also generated rationales aligned with the scoring rubric and provided actionable feedback for improvement. Third, increasing research has examined the stance-taking and rhetorical presentation of evaluative feedback produced by LLMs. For example, Kashiha (2025) compared comments produced by ChatGPT and human instructors on academic introductions written by Malaysian L2 students. The study found that both relied heavily on mitigation. However, human instructors were more assertive and more likely to use dialogic feedback to guide revisions, whereas ChatGPT tended to provide monologic suggestions that were efficient but relatively static and less interactive.

2.3. Chinese L2 AWE

Although AWE for native Chinese writing has made some progress (Yang et al., 2023), Chinese L2 AWE remains at an early stage and has primarily focused on automated scoring tasks. Existing studies indicate that Chinese L2 AWE has largely followed the developmental trajectory of English L2 AWE, progressing through three stages. Early work relied on machine learning with hand-crafted features, in which lexical, syntactic, and discourse features were extracted via feature engineering and modeled using traditional algorithms (Wang & Hu, 2021). This was followed by a shift to deep learning, with neural networks automatically learning latent semantic and structural representations, reducing dependence on manual features (Xiao et al., 2020). More recently, research has explored pretrained language models such as BERT, which have yielded significant improvements in both scoring consistency and validity (Rao & He, 2024).

However, while LLM-based AWE has been extensively explored for English L2 writing, research on Chinese L2 AWE remains at an early stage. As one of the earlier attempts, Poole and Coss (2024) explored the potential application of LLMs in low-stakes classroom assessment of Chinese L2 writing and found that ChatGPT can serve as a valuable tool in L2 writing assessment, provided that carefully designed prompting strategies are employed. Yang et al. (2026) fine-tuned Qwen3-8B on approximately 10,000 essays written by Chinese L2 learners and reported substantial alignment with human ratings, as evidenced by QWK = 0.7969, Spearman's ρ = 0.8010, and Pearson's r = 0.8023. However, this study did not incorporate a rubric-based scoring procedure, nor did it adopt a multi-agent framework to simulate multi-rater scoring and adjudication processes in authentic assessment settings. Cheng and Li (2025) made an initial attempt at multi-agent automated essay scoring for Chinese L2 writing and found that collaborative agents not only maintained scoring accuracy but also offered advantages in interpretability and multidimensional validity. However, in their approach, the agents in the first stage were assigned to evaluate separate dimensions of writing rather than making holistic judgments on the entire essay, as human raters typically do in authentic assessments. This approach may lead to computational redundancy and diverge from the logic underlying authentic writing assessment. In addition, preliminary evidence suggests that LLMs can support Chinese L2 learners in improving their writing performance (Song & Tang, 2025), further underscoring their value as a formative feedback supplement within AWE. Yang and Chen (2025) showed that ChatGPT-4.0 demonstrated superior performance in generating more accurate, detailed, and understandable corrective feedback compared to ChatGPT-3.5, highlighting its potential as a valuable AWE tool for Chinese L2 learners. However, to date, relatively few studies have examined the dual capability of LLMs in scoring and feedback generation for Chinese L2 writing.

2.4. Research gaps and questions

Despite the considerable progress made in previous LLM-based AWE studies, several limitations remain evident. First, the exploration of the potential of LLMs in Chinese L2 AWE is still limited. Existing LLM-based research on L2 writing has focused predominantly on English AWE, and linguistic indices are often treated as post hoc predictors rather than as construct-relevant evidence that can directly guide model reasoning. However, Chinese exhibits distinctive typological characteristics in its logographic writing system, morphology, and syntax (Li & Thompson, 1981), which makes it difficult to directly transfer English LLM-AWE frameworks or linguistic feature systems to the Chinese context without adaptation. Accordingly, there is a need to develop tailored solutions that reflect the specific characteristics of Chinese and to examine whether systematically embedding multidimensional, Chinese-specific linguistic indices into prompts can improve both score prediction and feedback generation. Second, current studies generally rely on single-agent scoring, which may introduce subjectivity and score-level bias, and fail to simulate the multi-rater collaboration and adjudication mechanisms used in authentic L2 writing assessment to enhance scoring consistency. This concern is reinforced by recent evidence of score-level bias in LLM-based scoring. AI raters have been found to avoid extreme scores (Tate et al., 2024), show inconsistent interpretation of adjacent lower-level ratings (Gjorevski et al., 2025), assign systematically higher scores to TOEFL essays (Liu et al., 2025b), or display conservative tendencies by being less likely than human raters and feature-based AES systems to assign high scores (Huang et al., 2026). These findings indicate that single-agent LLM scoring may be affected by systematic score-level biases. Therefore, beyond improving scoring consistency through multi-agent adjudication, it is also important to examine whether such a framework can reduce dependence on the scoring tendency of any single model and generate more stable and balanced score predictions across proficiency levels. Third, although prior studies have shown that fine-tuning can improve LLM performance in L2 AWE (Liu et al., 2025a, 2025b), fine-tuning research for Chinese L2 writing remains relatively scarce. Moreover, existing efforts rarely integrate construct-relevant linguistic evidence systematically into the fine-tuning process, and they seldom treat scoring and feedback as an integrated task for joint fine-tuning.

To address these limitations, this study proposes LAWE-CL2, a multi-agent LLM-based approach that integrates multidimensional

linguistic features, multi-agent adjudication, and fine-tuning. First, we construct a multidimensional system of linguistic indices covering lexical complexity, syntactic complexity, discourse cohesion, and linguistic accuracy. These indices are aligned with the HSK writing rubric to design refined prompting strategies and an automated processing pipeline. Second, we introduce a multi-agent framework to simulate collaborative rating scenarios, in which two primary agents generate initial evaluations and one meta agent adjudicates the final score to enhance fairness and stability. Finally, we fine-tune smaller-scale LLMs to assess the effect of fine-tuning on both scoring performance and feedback quality. The study is guided by three research questions:

RQ1: Can the integration of multidimensional linguistic features improve the performance of LLMs in Chinese L2 AWE?

RQ2: Can a multi-agent scoring framework achieve better performance in Chinese L2 AWE compared with a single-agent approach?

RQ3: Can fine-tuning further improve the performance of LLMs in Chinese L2 AWE?

3. Method

AWE research should employ a systematic validity framework to evaluate the appropriateness of automated assessment tools (Williamson et al., 2012; Wilson & Roscoe, 2020). We base our inquiry on the framework proposed by Williamson et al. (2012) for evaluating and implementing automated scoring systems for constructed-response writing tasks, which is theoretically grounded in Kane's argument-based validity model (Kane, 1992, 2006). Following Gjorevski et al. (2025), we focus on construct relevance and representation, as well as empirical performance, which is operationalized as agreement with human ratings. To support construct relevance and representation, we first anchor the target construct in the HSK writing rubric and operational rating procedures, and then operationalize the construct through a multidimensional, empirically validated set of linguistic indices covering lexical complexity, syntactic complexity, discourse cohesion, and linguistic accuracy. We screen and refine these indices based on their associations with score levels and potential redundancy, and further derive level-specific benchmark distributions from the training data. These construct-relevant signals are then systematically embedded into our advanced prompts so that scoring and feedback generated by the LLM can be grounded in explicit, interpretable evidence rather than impressionistic judgments. To support empirical performance, we evaluate the system's score outputs against human ratings through a set of complementary agreement and performance metrics, together with a series of controlled comparative settings. The specific metrics are introduced in Section 3.6. In validating and examining LLM-generated evaluation feedback, we draw on the work of Poole and Coss (2024), Saricaoglu and Bilki (2025), and Gjorevski et al. (2025). The specific procedures are described in the following sections.

3.1. Data source and the scoring rubric

The corpus used in this study is the HSK Dynamic Composition Corpus, which comprises essay responses produced by international learners of Chinese (non-native speakers) for the writing task of the old version of the Advanced HSK⁹ between 1992 and 2005. It is a large-scale, publicly accessible collection of Chinese L2 learner essays maintained by Beijing Language and Culture University. The exam was time-limited, and according to the official test syllabus, examinees were required to complete the writing task within 30 min. The writing task adopts a constrained, prompt-based format. In each test administration, examinees were required to produce only one essay. The responses were handwritten, and examinees were not allowed to use any external aids during the writing process. This corpus contains 11,337 essays totaling 3715,987 Chinese characters.

The corpus further includes multidimensional error annotations at five levels (character, word, sentence, paragraph, and punctuation). These error tags were officially developed for the HSK and manually annotated by trained HSK raters in accordance with the official annotation guidelines. In total, the corpus contains 52 error-tag categories and 310,550 annotated instances. Yan and Lin (2023) further validated the quality of the HSK error-tag annotations by double-coding 100 essays and comparing their coding with the corpus tags, reporting near-perfect agreement ($K = .978$, $p < .001$). The corpus encompasses a variety of genres (e.g., argumentative and narrative writing) and reflects diverse native language backgrounds, ensuring topic diversity and enhancing the generalizability of the research findings (see Appendix A for details). Previous research has evaluated the HSK exam using the Assessment Use Argument framework, indicating that it is a well-structured and reliable exam (Peng et al., 2020).

HSK writing is scored through standardized procedures by raters with master's or doctoral degrees and experience in teaching Chinese as a second language. Raters first undergo rigorous training on the scoring rubric. Each essay is then independently scored by two raters. When the two ratings differ by no more than two points, the interrater agreement is considered acceptable, and the final score is calculated as the average of the two ratings (Yang et al., 2023). When the two ratings differ by more than two points, a third rater adjudicates to determine the final score. According to the official HSK Scoring Guidelines,¹⁰ essays are classified into three levels (low, medium, high), with blank essays assigned a score of zero. The HSK writing test primarily adopts a holistic scoring method supplemented by a five-level analytic rubric, which provides qualitative descriptors to guide raters in forming a comprehensive overall impression but does not assign numerical values to individual categories (Nie, 2009). HSK essay scores range from 40 to 95 in 5-point increments. Following previous research (Huang et al., 2023; Hao et al., 2026), this study treats scores of 40–59 as low level (C) essays, 60–79 as medium level (B) essays, and 80–100 as high level (A) essays. The detailed rubric is provided in Appendix B. All essays were drawn from one of the 25 writing topics listed in Appendix A and were distributed across three genres (argumentative, narrative, and practical writing) as shown in Table 1. Because this study involves training and testing LLMs, the collected dataset is divided into a

⁹ HSK corpus (3.0): <https://hsk.blcu.edu.cn/> (accessed 5 May 2025)

¹⁰ https://www.chinesetest.cn/userfiles/maket_download/2015/no4_HSK.pdf (accessed 10 September 2025)

Table 1

The details of the corpus used in this study.

Level		low (C)	medium (B)	high (A)
Essays (<i>N</i>)		2000	2000	2000
Tokens		320,428	443,903	504,970
Sentences		21,732	29,369	27,714
Genres	Argumentative	81.50%	75.45%	65.20%
(Proportion %)	Narrative	9.85%	13.10%	24.35%
	Practical	8.65%	11.45%	10.45%
Top 5 Countries		Korea	Korea	Singapore (16.90%)
(Proportion %)		(47.30%)	(37.20%)	Japan
		Japan	Japan	(15.25%)
		(29.20%)	(31.95%)	Korea
		Indonesia	Singapore	(14.05%)
		(4.15%)	(7.15%)	Malaysia
		Thailand	Indonesia	(11.90%)
		(3.25%)	(5.65%)	Indonesia (10.65%)
		Singapore	Malaysia	
		(2.10%)	(2.60%)	

training set (80%) and a test set (20%). The training set comprises 4800 essays, with 1600 essays per proficiency level, while the test set comprises 1200 essays, with 400 essays per level. Both the training and test sets contain the essay text, the corresponding error annotations, and the original scores assigned by HSK raters. The distribution of essay levels is balanced in both sets, enabling LLMs to be trained and tested fairly.

3.2. Calculation of linguistic indices in Chinese L2 Essays

L2 learner language production is a multidimensional construct involving lexical, syntactic, and discourse skills (Lu, 2025). Accordingly, writing proficiency is often operationalized using quantitative indices of complexity and accuracy (Bulté & Housen, 2012). We used the HSK rubric and prior research on Chinese L2 writing complexity to select validated linguistic indices predictive of writing quality. These indices span four dimensions of writing, namely lexical complexity (Hao et al., 2023a; Yan & Lin, 2023; Zhang & Lei, 2025), syntactic complexity (Hu et al., 2022; Hao et al., 2022; Hao et al., 2023b; Lu & Wu, 2022; Hao et al., 2024; Hao et al., 2025; Wang et al., 2025), discourse cohesion (Peng et al., 2023), and linguistic accuracy (Yan & Lin, 2023). The selected indices were calculated using self-developed Python scripts and L2C-SCA (Bin et al., 2025). For linguistic accuracy, the indices were computed based on the error annotation set in the HSK corpus, following Yang and Chen (2025) and Yang et al. (2026). Pearson correlation coefficients were used to examine the relationship between linguistic complexity indices and Chinese L2 writing proficiency levels (Levels 1–3). Of the 69 linguistic indices initially selected, 9 indices were excluded due to non-significant correlations ($ps > 0.05$), and 8 indices were removed due to multicollinearity (Pearson correlation greater than 0.70 with other indices), leaving a final set of 52 linguistic indices (see Appendix C). Table 2 presents sample indices from the lexical, syntactic, cohesion, and accuracy dimensions, while the complete set of 52 indices with descriptive statistics is provided in Appendix D. Subsequently, we conducted a linguistic empirical analysis of these indices by drawing on previous research. Finally, we transformed these empirical analyses into prompts designed to guide LLMs in better aligning with the construct representation of Chinese L2 writing during automated scoring and feedback generation (see Appendix E for details).

3.3. Prompt design and baseline LLMs

Following Liu et al. (2025a) and Liu et al. (2025b), we reclassified the original five-level HSK writing rubric into three levels. Specifically, Levels 1–2 were merged into Level C, Level 3 was retained as Level B, and Levels 4–5 were combined into Level A. This procedure is consistent with the method described in Section 3.1 for aggregating raw essay scores. Additionally, building on Li and Liu (2024), this study represents the first systematic attempt to integrate a multidimensional set of linguistic indices into Chinese L2 writing assessment. We not only explicate the construct meaning of these linguistic indices and their correlations with writing scores across score levels, but also provide benchmark distributions derived from the standardized data, operationalized as the mean values for Levels A, B, and C. This design enables the model to align its scoring with key constructs of Chinese writing rather than relying solely on holistic impressions. We design an automated pipeline in which, once an essay is submitted, the system computes its linguistic feature data by running compiled programs. The essay text and its corresponding linguistic indices data are then embedded into a pre-constructed prompt template, which is subsequently fed into the LLM. This enables a fully automated process from essay input to the generation of both scores and feedback.

We design two prompt templates to compare the role of linguistic features in LLM-based automated scoring (see Appendix E). The

Table 2

Linguistic indices values (samples) across three levels of L2 Chinese writing.

sample indices	dimension	low (C)	medium (B)	high (A)
1. NDW	lexical	96.715	125.773	149.041
2. Uber	lexical	51.473	52.806	59.639
.....				
17. MLS	syntactic	26.437	27.694	32.991
18. MLC	syntactic	11.706	11.394	11.116
.....				
41. pron_density	cohesion	0.104	0.105	0.095
42. conj_density	cohesion	0.044	0.044	0.042
.....				
50. norm_chi_word_errors	accuracy	3.401	2.627	1.953
51. norm_sentence_errors	accuracy	2.184	1.429	0.803
.....				

Note. The indices shown here are illustrative samples. Full definitions and computation formulas for all indices are provided in Appendix C.

first general prompt includes only the HSK writing rubric and is intended to examine the model's performance when relying solely on holistic impressions of the essay without additional linguistic features. The second advanced prompt extends the scoring rubric by incorporating 52 empirically validated linguistic indices together with their benchmark distributions and explanatory analyses, enabling the model to account for developmental patterns across proficiency levels and to generate more substantiated scoring and feedback.¹¹ Li and Liu (2024) found that GPT-4 produced consistent scoring outcomes regardless of the prompt language. Building on this finding, this study adopt Chinese prompts across all template designs to better reflect the authentic context of Chinese L2 writing assessment. In terms of output, our prompts require the LLMs to produce results in a uniform JSON format, reporting both the final writing level and feedback across five dimensions (content and communicative function, vocabulary, syntax, discourse cohesion, and linguistic accuracy).

For the purpose of selecting appropriate baseline LLMs to perform the AWE task, we estimate the response burden associated with the two prompting strategies. Specifically, we use the tiktoken¹² function released by OpenAI to calculate the number of input tokens¹³ generated by the two prompting templates when applied to different essays. We find that for the general prompt, the maximum number of input tokens is 1864, the minimum is 874, and the median is 1299. For the advanced prompt incorporating a scoring rubric and linguistic indices, the maximum is 11,679, the minimum is 10,639, and the median is 11,111. Accordingly, it is essential to employ LLMs with state-of-the-art Chinese processing capabilities for handling long contexts. Thus, GPT-4.1-mini, DeepSeek-Chat-V3, and Qwen3-Next-80B-A3B-instruct (Qwen3-80B) are selected as the baseline models in our experiments. By comparing these three LLMs across the two prompting strategies, we systematically assess not only the feasibility of handling different levels of token burden but also the reliability and validity of automated writing evaluation under varying prompt complexities. Moreover, these LLMs can be conveniently accessed via APIs. In this paper, GPT-4.1-mini is accessed through the OpenAI platform,¹⁴ DeepSeek-Chat-V3 through the DeepSeek platform,¹⁵ and Qwen3-80B through the OpenRouter platform.¹⁶

3.4. Multi-agent scoring framework

In operational writing assessment contexts, essay evaluation often involves collecting independent scores from multiple raters before any adjudication or score resolution takes place. For example, TOEFL writing assessment uses a mechanism in which both the e-rater system and a human rater provide scores (Enright & Quinlan, 2010). If there is a substantial discrepancy between the human rater and the e-rater, a second human rater is invited to provide an additional score. Such collaborative scoring practices are also found in the HSK writing assessment, even though no machine raters are involved. However, it is important to note that collaborative scoring in operational assessment settings usually means that each rater performs the same scoring task independently, for instance by assigning scores and providing evaluative comments based on the same set of linguistic perspectives. Inspired by this approach, we develop a multi-agent scoring framework to simulate a multi-rater collaborative scoring mechanism. Specifically, two distinct agents function as primary raters, evaluating the writing based on the same prompt and providing corresponding feedback independently. Subsequently, a meta agent serves as an arbitrator, drawing on the feedback from both primary agents while incorporating its own judgment of the essay to make the final decisions. This meta agent is responsible for delivering the final score level as well as the accompanying feedback.

¹¹ In this study, feedback consists of the rationales of the assigned score, the identification of shortcomings across multiple dimensions, and suggestions for revision. These components are generated in an integrated manner rather than as isolated elements.

¹² <https://github.com/openai/tiktoken> (accessed 10 June 2025)

¹³ The token here refers to the number of tokens after word segmentation by the exclusive tokenization module of the LLM, which is usually greater than the number of Chinese words in the text.

¹⁴ <https://openai.com/api/> (accessed 10 June 2025)

¹⁵ <https://api-docs.deepseek.com/> (accessed 10 June 2025)

¹⁶ <https://openrouter.ai/> (accessed 10 June 2025)

To ensure the effective execution of the meta agent, we design a dedicated meta prompt built upon the advanced prompt. This meta prompt provides more comprehensive information, including the scoring rubric, linguistic features, essay content, and feedback from the two primary raters (see Appendix F for details). Although appending additional feedback increases the context length, it helps the meta agent draw on more evidence when making its final decision. Overall, we suggest that the multi-agent scoring framework may provide two potential advantages. First, it helps mitigate the subjectivity and bias associated with reliance on a single rater. Second, it may reduce the complexity and resource costs associated with assigning different raters to perform separate tasks on the same essay. During inference, GPT-4.1-mini, DeepSeek-Chat-V3, and Qwen3-Next-80B-A3B-instruct are alternately employed to act as different agents. For instance, GPT-4.1-mini and DeepSeek-Chat-V3 function as the two primary agents, while Qwen3-Next-80B-A3B-instruct serves as the meta agent.

3.5. Synthetic feedback data generation and fine-tuning

The training data consists only of essays, error annotations, and original scores annotated by HSK experts. Therefore, it is insufficient for fine-tuning a unified AWE system for both scoring and feedback generation, as it lacks target feedback. To address this issue, we first create a new prompt by integrating essays, original scores, the scoring rubric, and linguistic features, which is then used to instruct LLMs to generate synthetic feedback (see Appendix G for details). Compared with the advanced prompt that incorporates the scoring rubric and linguistic features, this version additionally makes the original essay scores visible to the LLMs. Specifically, we employ Qwen3-80B to generate feedback for 4800 training essays. The model is required to provide rationales for why each essay receives a particular score level and to identify in detail its shortcomings across multiple dimensions, along with suggestions for improvement. Following this step, we randomly sample 400 instances for each level. Then, a manual verification was conducted, in which two doctoral students with backgrounds in L2 writing assessment independently evaluated whether the generated evaluation feedback accurately explained both the essay content and its assigned score. The Cohen's Kappa agreement of these two annotators is 0.82 and the proportion of valid annotations is 91%, indicating that the synthetic data are both reliable and valuable. Finally, we calculate the token length of the synthetic feedback using the tiktoken function. The maximum length is 4928 tokens, the minimum 2224 tokens, and the median 3289 tokens, highlighting that the response burden of the AWE task is considerably greater than that of traditional text classification tasks.

High-quality instruction data have been shown to enhance the domain-specific capabilities of LLMs in various areas such as linguistics and education (Yang et al., 2025; Yang et al., 2026). LLMs with relatively smaller parameter scales (e.g., 7B or 8B) can achieve competitive performance comparable to their ultra-large counterparts (e.g., 236B) after fine-tuning. Accordingly, we fine-tune Qwen3-8B and GPT-4.1-nano-2025-04-14 (GPT-4.1-nano)¹⁷ using the training data consisting of augmented 4800 essays with original scores and synthetic feedback. In this paper, the experiment for fine-tuning Qwen3-8B is implemented on Pytorch 2.6.0 and 6 NVIDIA A100-PCIE (40 GB) GPUs using LLaMA-Factory. LoRA is utilized to reduce computing costs during fine-tuning, where the learning rate is 5e-5, the epoch is 3, the batch size is 1, and bf16 is used as the compute type. The total time for fine-tuning Qwen3-8B is approximately 6 h. In addition, the fine-tuning experiment for GPT-4.1-nano is conducted on the OpenAI platform. We convert the training data into the format of OpenAI instruction data and then upload it to the platform, where a fine-tuning task is created and execute automatically. The fine-tuning hyperparameters are also set automatically by the OpenAI platform, with the number of epochs set to 3, the learning rate to 0.1, and the batch size to 1. The total time for fine-tuning GPT-4.1-nano is about 160 min. During fine-tuning, 10% of the training data is used as the validation set to track the learning effectiveness of these two LLMs. The validation loss of fine-tuning Qwen3-8B is presented in Fig. 1. It shows that the loss drops from 1.201 to 0.372 and flattens out after 1600 steps.

Similarly, the validation loss of fine-tuning GPT-4.1-nano is reported in Fig. 2. It shows that the loss drops from 1.498 to 0.509 and flattens out after 8000 steps. These results indicate that both Qwen3-8B and GPT-4.1-nano can acquire domain-specific knowledge to improve their abilities for assessing writing through fine-tuning. Eventually, the fine-tuned Qwen3-8B and GPT-4.1-nano are set as the primary agents to assess writing independently, while Qwen3-80B is set as the meta agent to make final decisions based on the feedback from Qwen3-8B and GPT-4.1-nano. Together, they constitute the final components of LAWE-CL2.

3.6. Metrics for empirical performance

This study requires the AWE system to output both scores and feedback in JSON format. Pilot experiments show that general-purpose LLMs sometimes produce garbled text or repeated fields, failing to meet specifications. We regard such cases as task incompleteness and adopt task completeness as an evaluation metric, since robust AWE systems must achieve high completeness to ensure satisfactory real-world applications. To evaluate the consistency between system predictions and human ratings, we use two common AWE metrics, quadratic weighted kappa (QWK) and exact agreement rate (EAR), which are also used in assessing the reliability of LLM-based AWE (Williamson et al., 2012; Sun et al., 2025; Liu et al., 2025a). Higher QWK and EAR values, with values closer to 1, indicate greater agreement between system predictions and human ratings. In this study, since the system is required to predict discrete A-B-C levels, EAR can also be interpreted as an accuracy measure. In addition, we use Pearson's correlation coefficient, Spearman's rank correlation, and the F1-score to evaluate the model's performance in predicting the three proficiency levels.

To validate feedback generation quality, we integrate the approaches of Poole and Coss (2024), Saricaoglu and Bilki (2025), and

¹⁷ These two LLMs were selected because they had great and robust instruction-following abilities in our pilot explorations and they had relatively fast speed during inference.

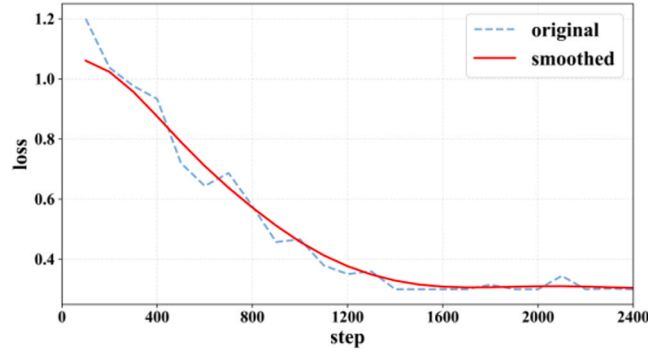


Fig. 1. Loss of fine-tuning Qwen3-8B in the validation set.

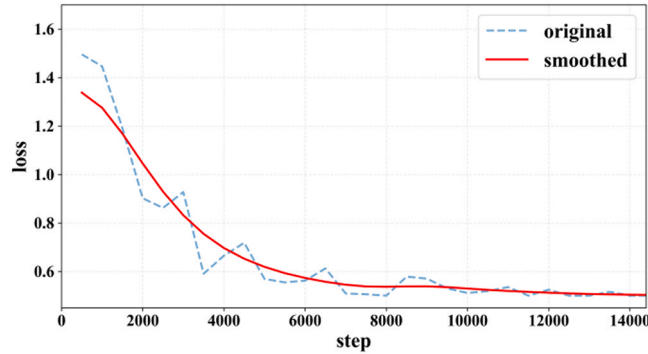


Fig. 2. Loss of fine-tuning GPT-4.1-nano in the validation set.

Gjorevski et al. (2025) and adopt a hybrid evaluation strategy that encompasses three dimensions, including coverage of objective evidence, internal consistency, and overall feedback quality. First, coverage of objective evidence primarily assesses whether the model's scoring rationale is grounded in verifiable evidence and whether statistical statements are accurate. Second, internal consistency evaluates the logical coherence within each comment or section of feedback, the consistency of terminology used, and the alignment between evidence and conclusions. Third, following the approach of Gjorevski et al. (2025), we evaluate the overall feedback quality from three aspects of tone, text specificity, and mitigation strategies. These three dimensions are consolidated into a new prompt (see Appendix H for details), which instructs the model to assign a score from 1 to 5 for each dimension, with higher scores indicating better quality. This prompt is used to instruct Qwen3-80B to automatically examine all feedback generated in our experiments, thereby reducing the burden on human annotators. Finally, to validate the reliability of this automated approach for examining evaluation feedback generation, we randomly sample 200 outputs from each of the three baseline LLMs on the test set and conduct human verification. Between Qwen3-80B and two annotators, the Cronbach's alpha of the coverage of objective evidence, internal consistency of feedback and overall feedback quality were 0.844, 0.820 and 0.816 respectively. These results demonstrate that this approach is both effective and trustworthy, making it suitable for large-scale automated evaluation.

3.7. The settings for comparative experiments

To evaluate the effectiveness of incorporating multidimensional linguistic features and their detailed analytical information into scoring prompts, as well as the contributions of the multi-agent scoring framework and fine-tuning, we conduct five comparative experimental settings, summarized in Table 3. Settings 1 and 2 compare a general prompt and an advanced prompt to examine the added value of multidimensional linguistic features. Setting 3 evaluates the effectiveness of a multi-agent AWE system. Setting 4 evaluates the effectiveness of fine-tuning by using LLMs fine-tuned on synthetic feedback data. Setting 5 evaluates the effectiveness of LAWE-CL2, which integrates multidimensional linguistic features, multi-agent collaboration, and fine-tuning.

4. Results

The main results of the five comparative experiments are summarized in Table 4. To maintain a clear link between our analyses and the research questions, we organize the Results around RQ1-RQ3. Specifically, RQ1 is addressed by comparing the single-agent zero-shot baselines with the general versus advanced prompt (Settings 1 and 2), RQ2 by contrasting the multi-agent framework against single-agent baselines under matched prompting conditions (Setting 3 versus Settings 1–2), and RQ3 by examining whether fine-

Table 3
Comparative experimental settings for LAWE-CL2.

Setting	Experimental focus	Models	Prompts	Multi-agent Scoring	Fine-tuning
1	Performance of baseline LLMs on Chinese L2 AWE	GPT-4.1-mini; DeepSeek-Chat-V3; Qwen3-80B	General prompt	No	No
2	Performance of baseline LLMs on Chinese L2 AWE with multidimensional linguistic features	GPT-4.1-mini; DeepSeek-Chat-V3; Qwen3-80B	Advanced prompt	No	No
3	Performance of multi-agent collaborative scoring	GPT-4.1-mini; DeepSeek-Chat-V3; Qwen3-80B	Advanced prompt + meta prompt	Yes	No
4	Performance of fine-tuned LLMs	GPT-4.1-nano; Qwen3-8B	advanced prompt	No	Yes
5	Final performance of LAWE-CL2	Primary: fine-tuned GPT-4.1-nano & Qwen3-8B; Meta: Qwen3-80B	advanced prompt + meta prompt	Yes	Yes

Note: across all experiments, we fixed the temperature at 0.1 and set the maximum output length to 4096 tokens.

tuning further improves performance and robustness (Settings 4–5).

4.1. RQ1: The effectiveness of multidimensional linguistic features (Settings 1 and 2)

To answer RQ1, we conduct experiments under Settings 1 and 2. In Setting 1, as shown in Table 4, when the LLMs are provided only with the HSK scoring rubric through the general prompt, all three baseline models demonstrate relatively low performance in both scoring and feedback. For scoring, taking QWK as an example, DeepSeek-Chat-V3 achieves only 0.3209, and GPT-4.1-mini achieves 0.3529. Although Qwen3-80B reaches a QWK of 0.5066, its EAR is only 0.5458, indicating that it predicts the correct scores for only about half of the test set. For the evaluation feedback, both DeepSeek-Chat-V3 and GPT-4.1-mini achieve objective evidence coverage scores below 3 and overall feedback quality scores of only about 3.7. These are the lowest values among the five experimental settings, highlighting the limitations of the general prompt. However, when the advanced prompt is used, Setting 2 demonstrates better performance than Setting 1. For instance, compared with Setting 1, Setting 2 yields notable improvements. In absolute terms, QWK improves by + 0.1847 for DeepSeek-Chat-V3, + 0.1913 for GPT-4.1-mini, and + 0.1033 for Qwen3-80B. Likewise, the F1-score increases by + 0.1365, + 0.1418, and + 0.0687 for the three models, respectively. It is worth noting that all three LLMs show substantial improvements across the three dimensions of evaluation feedback effectiveness, with all scores exceeding 4.5. In terms of objective evidence coverage, DeepSeek-Chat-V3 increases from 2.8555 to 4.6030, GPT-4.1-mini rises from 2.9833 to 4.8325, and Qwen3-80B improves from 4.3700 to 4.9358. For both scoring and evaluation feedback, the advanced prompt yields consistent performance improvements across all three LLMs, suggesting that the effects are robust across models. These findings address RQ1. The results show that incorporating multidimensional linguistic features, together with benchmark distributions and explanatory analyses, into the scoring rubric improves the effectiveness of LLM-based AWE. This integration provides richer evidence to support more accurate score predictions, and it also enables the models to generate higher-quality evaluation feedback grounded in the same linguistic evidence.

In the single-agent experiments of Settings 1 and 2, we find that DeepSeek-Chat-V3 performs well on the adjacent agreement metric, with values above 0.99 in both settings. This indicates that it is less likely to produce non-adjacent misclassifications, such as predicting an essay as Level A when its true level is C and skipping Level B. However, across the evaluation metrics for both scoring and feedback, DeepSeek-Chat-V3 demonstrates the lowest overall performance. Therefore, in the subsequent single-agent fine-tuning experiments, we select relatively small-scale models from the GPT and Qwen series as the backbone models. In addition, the completion rates of the LLMs remain above 99% in all experiments, highlighting the feasibility of the advanced prompt. Although the response burden is heavier than that of traditional text classification tasks, the selected LLMs still accomplish the AWE tasks effectively and generate outputs as required.

4.2. RQ2: The effectiveness of the multi-agent scoring framework (Setting 3)

To address RQ2, we conduct the experiments under Setting 3. Results show that all three multi-agent setups outperform their respective single-agent versions in Setting 2 in terms of both scoring prediction and feedback generation. Specifically, taking QWK and F1 as examples, when DeepSeek-Chat-V3 is designated as the meta agent, the QWK shows an absolute increase of + 0.1694 and the F1-score shows an absolute increase of + 0.1367 compared with its single-agent performance. Similarly, when GPT-4.1-mini serves as the meta agent, its QWK shows an absolute increase of + 0.1064 and the F1-score shows an absolute increase of + 0.1031. When Qwen3-80B acts as the meta agent, the system achieves an absolute increase of + 0.1054 in QWK and an absolute increase of + 0.0919 in F1-score compared with Qwen3-80B functioning as a single agent. In addition, its EAR approaches 0.7000, indicating that the multi-agent configuration predicts essay score levels correctly for about 10% more cases than the single-agent setup. In terms of evaluation feedback effectiveness, all three multi-agent frameworks score above 4.5 across the dimensions of objective evidence coverage, internal consistency, and overall feedback quality, and each shows small but consistent improvements compared with its respective single-agent settings.

These findings answer RQ2 and suggest that multi-agent collaboration not only enhances the accuracy of Chinese L2 writing assessment but also produces more reliable feedback, thereby further activating the potential of LLMs. In summary, these findings

Table 4
Main results of the five comparative experiments.

Models	Completion Rate	Scoring Performance						Feedback Performance		
		QWK	EAR	Adjacent Agreement	Spearman ρ	Pearson r	F1-score	Objective Evidence Coverage	Internal Consistency	Overall Feedback Quality
Setting 1: General-purpose LLMs using the general prompt										
DeepSeek-Chat-V3	0.9783	0.3209	0.4472	0.9966	0.4026	0.4022	0.3949	2.8555	3.9198	3.7886
GPT-4.1-mini	1.0000	0.3529	0.4583	0.9942	0.4374	0.4372	0.4053	2.9833	4.0800	3.7033
Qwen3-80B	1.0000	0.5066	0.5458	0.9750	0.5326	0.5349	0.5427	4.3700	4.8800	4.5442
Setting 2: General-purpose LLMs using the advanced prompt										
DeepSeek-Chat-V3	0.9767	0.5056	0.5401	0.9923	0.5445	0.5451	0.5314	4.6030	4.6722	4.5780
GPT-4.1-mini	0.9900	0.5442	0.5522	0.9874	0.5814	0.5811	0.5471	4.8325	4.8400	4.5567
Qwen3-80B	1.0000	0.6099	0.6033	0.9783	0.6149	0.6149	0.6114	4.9358	4.9450	4.9525
Setting 3: Multi-agent systems using the advanced prompt + meta prompt										
GPT-4.1-mini + Qwen3-80B -> DeepSeek-Chat-V3 (meta agent)	0.9900	0.6750	0.6608	0.9874	0.6829	0.6830	0.6681	4.6914	4.6931	4.6030
DeepSeek-Chat-V3 + Qwen3-80B -> GPT-4.1-mini (meta agent)	0.9925	0.6506	0.6448	0.9924	0.6695	0.6694	0.6502	4.8908	4.9083	4.8758
DeepSeek-Chat-V3 + GPT-4.1-mini -> Qwen3-80B (meta agent)	0.9950	0.7153	0.6960	0.9925	0.7251	0.7259	0.7033	4.9775	4.9842	4.9817
Setting 4: Fine-tuned LLMs using the advanced prompt										
Fine-tuned GPT-4.1-nano	1.0000	0.7109	0.7544	0.9666	0.7154	0.7157	0.7566	4.8450	4.8392	4.8492
Fine-tuned Qwen3-8B	1.0000	0.7849	0.7993	0.9783	0.7873	0.7876	0.8020	4.9633	4.9633	4.9650
Setting 5: Fine-tuned multi-agent system using the advanced prompt + meta prompt										
Fine-tuned GPT-4.1-nano + Fine-tuned Qwen3-8B -> Qwen3-80B (meta agent)	1.0000	0.7989	0.8125	0.9792	0.8010	0.8011	0.8149	4.9817	4.9825	4.9808

provide empirical support for the effectiveness of a multi-agent scoring framework in Chinese L2 AWE. The performance gains can be attributed to the meta agent's ability to integrate the prior feedback generated by the two primary agents, which effectively provides it with additional two-shot examples. This process enables the meta agent to engage in more effective in-context learning and to synthesize multiple perspectives before making the final judgment. In Settings 1 and 2, Qwen3-80B already exhibits the strongest baseline performance. Accordingly, when it serves as the meta agent, the corresponding multi-agent configuration also achieves the best performance. This outcome can be explained by its ability not only to incorporate the two prior evaluations provided by the primary agents but also to integrate its own inherent evaluative strengths in making the final decision.

To further evaluate empirical performance in terms of reliability and fairness (Williamson et al., 2012), we conduct supplementary experiments to examine internal consistency among agents and whether the meta agent systematically favors either of the primary agent. We employed Cronbach's alpha to calculate the internal consistency between rater 1¹⁸ and rater 2,¹⁹ between rater 1 and the meta agent, between rater 2 and the meta agent, as well as among all three collectively. The results are presented in Table 5. First, the consistency between rater 1 and rater 2 reaches 0.8988, indicating good agreement between the fine-tuned GPT-4.1-nano and Qwen3-8B. Second, the consistencies between rater 1 and the meta agent and between rater 2 and the meta agent are 0.9561 and 0.9339, respectively. The values both exceed 0.90, reflecting excellent agreement between the meta agent and the two fine-tuned LLMs. The small difference between them further suggests that the meta agent does not exhibit a strong bias toward either side. Third, the joint consistency among all three reaches 0.9526, which is 0.0538 higher than the agreement between rater 1 and rater 2 alone. This indicates that the inclusion of the meta agent enhances the overall coherence of the system, demonstrating that the multi-agent framework yields a consistency gain.

To provide a more fine-grained evaluation of the multi-agent framework in relation to RQ2, we further conduct a level-specific analysis to examine whether multi-agent adjudication improves prediction performance across different proficiency levels. The results are reported in Table 6. We find that all models exhibit a stronger tendency to predict Level B. For example, in Setting 1, DeepSeek-Chat-V3 reaches an EAR of 0.9125 at Level B (with a prediction proportion of 83.38%), whereas its EAR is only 0.1550 at level C (6.73% predicted) and 24.50% at level A (9.89% predicted). Moreover, we observe that in Setting 2 the three LLMs produce more balanced prediction distributions across the score levels compared with Setting 1. This also partially mitigates the tendency to concentrate predictions in Level B, with all models showing increased prediction proportions for essays at Levels C and A. For example, in the case of GPT-4.1-mini, the EAR for Level C improves from 0.3725 to 0.5900, with the prediction proportion rising from 0.1600 to 0.2786; for Level A, the EAR increases from 0.1025 to 0.3075, while the prediction proportion grows from 3.75% to 12.04%. Similarly, the multi-agent system in Setting 3 further improves both the prediction proportions and EAR across score levels compared with Setting 2. For example, compared with DeepSeek-Chat-V3 as a single agent, the multi-agent configuration of "GPT-4.1-mini+Qwen3-80B→DeepSeek-Chat-V3 (meta agent)" shows changes in prediction proportions across all three levels. The proportion of Level C predictions increases from 18.94% to 25.70%, and the proportion of Level A predictions rises from 12.37% to 23.67%. In contrast, the proportion of Level B predictions decreases from 68.69% to 50.63%.

4.3. RQ3: The effectiveness of fine-tuning and full LAWE-CL2 (Settings 4 and 5)

To examine whether fine-tuning further improves Chinese L2 AWE performance, we conduct the Setting 4 and Setting 5 experiments (RQ3). In Settings 4 and 5, the AWE systems achieve a completion rate of 100%, indicating that fine-tuning enables the LLMs to avoid producing garbled or repetitive outputs for our AWE tasks. Notably, the fine-tuned GPT-4.1-nano and Qwen3-8B outperform both the single-agent tests in Setting 2 and the multi-agent tests in Setting 3. This is an encouraging result, as GPT-4.1-nano and Qwen3-8B are relatively small-scale models within the GPT and Qwen3 series. Nevertheless, through fine-tuning, they are able to achieve performance that substantially surpasses that of the baseline LLMs. For example, relative to GPT-4.1-mini in Setting 2, the fine-tuned GPT-4.1-nano shows absolute increases of + 0.1667 in QWK, + 0.2022 in EAR, and + 0.2095 in the F1 score. Similarly, the fine-tuned Qwen3-8B shows an absolute increase of + 0.1750 in QWK, an absolute increase of + 0.1960 in EAR, and an absolute increase of + 0.1906 in F1-score over Qwen3-80B in Setting 2. They achieve scores exceeding 4.8 in objective evidence coverage, internal consistency, and overall feedback quality, indicating that their feedback is more reliable. Compared with the results in Setting 3, the fine-tuned LLMs also obtain improvements of more than 10% in both EAR and F1-score.

These results provide an answer to RQ3 and demonstrate that fine-tuning is effective for AWE tasks, as it not only reduces the cost of API calls but also substantially enhances the performance of small-scale LLMs, thereby offering valuable resources for future on-device deployment. Moreover, the effectiveness of our synthetic feedback data are confirmed. Empirical evidence shows that these data improve both scoring and feedback performance by enabling LLMs to learn the detailed logic of scoring and reasoning, thereby leading to a more accurate alignment between scores and underlying rationales.

Moreover, Setting 5 integrates the fine-tuned GPT-4.1-nano and Qwen3-8B as primary agents with Qwen3-80B as the meta agent, yielding the strongest performance across all five comparative settings. The system attains a QWK of 0.7989, exceeding the minimally acceptable threshold of 0.70 suggested by Williamson et al. (2012) and indicating acceptable performance, along with an EAR of 0.8125 and an F1 score of 0.8149. The system achieves scores above 4.98 in objective evidence coverage, internal consistency, and overall feedback quality. These results suggest that it not only predicts 81.25% of essay levels correctly but also produces feedback of high quality and reliability. It is also the only system with an EAR exceeding 0.800, highlighting its potential for practical application.

¹⁸ Rater 1 refers to the fine-tuned GPT-4.1-nano.

¹⁹ Rater 2 refers to the fine-tuned Qwen3-8B.

Table 5

Internal consistency results of the multi-agent framework.

Models	Cronbach's α			
	R1-R2	R1-Meta	R2-Meta	R1-R2-Meta
Fine-tuned GPT-4.1-nano + Fine-tuned Qwen3-8B -> Qwen3-80B (meta agent)	0.8988	0.9561	0.9339	0.9526

Note: R1 refers to rater agent 1, R2 refers to rater agent 2, and meta refers to the meta agent.

Table 6

Chinese L2 AWE performance of the five experimental settings across three score levels.

Models	Scoring EAR			Prediction Proportion		
	low (C)	medium (B)	high (A)	low (C)	medium (B)	high (A)
Setting 1: General-purpose LLMs using the general prompt						
DeepSeek-Chat-V3	0.1550	0.9125	0.2450	0.0673	0.8338	0.0989
GPT-4.1-mini	0.3725	0.9000	0.1025	0.1600	0.8025	0.0375
Qwen3-80B	0.5500	0.7475	0.3400	0.2717	0.5942	0.1342
Setting 2: General-purpose LLMs using the advanced prompt						
DeepSeek-Chat-V3	0.4350	0.8350	0.3125	0.1894	0.6869	0.1237
GPT-4.1-mini	0.5900	0.7425	0.3075	0.2786	0.6010	0.1204
Qwen3-80B	0.5850	0.6625	0.5625	0.2517	0.4833	0.2650
Setting 3: Multi-agent systems using the advanced prompt + meta prompt						
GPT-4.1-mini + Qwen3-80B -> DeepSeek-Chat-V3 (meta agent)	0.6200	0.7625	0.5775	0.2570	0.5063	0.2367
DeepSeek-Chat-V3 + Qwen3-80B -> GPT-4.1-mini (meta agent)	0.5200	0.8500	0.5500	0.1998	0.5861	0.2141
DeepSeek-Chat-V3 + GPT-4.1-mini -> Qwen3-80B (meta agent)	0.6400	0.8300	0.6075	0.2605	0.5193	0.2203
Setting 4: Fine-tuned LLMs using the advanced prompt						
Fine-tuned GPT-4.1-nano	0.6825	0.8850	0.6900	0.2717	0.4700	0.2583
Fine-tuned Qwen3-8B	0.7625	0.8750	0.7525	0.2933	0.4317	0.2750
Setting 5: Fine-tuned multi-agent system using the advanced prompt + meta prompt						
Fine-tuned GPT-4.1-nano + Fine-tuned Qwen3-8B -> Qwen3-80B (meta agent)	0.7825	0.8850	0.7700	0.2692	0.4442	0.2867

We argue that the superior performance of LAWE-CL2 primarily stems from the informative and highly accurate prior feedback provided by the fine-tuned GPT-4.1-nano and Qwen3-8B, which enables Qwen3-80B, already strong in overall performance, to make final decisions with more effective reference information. Thus, fine-tuning can further amplify its contribution to multi-agent AWE systems through this form of collaborative scoring.

The analyses above show that the full LAWE-CL2 system achieves strong overall scoring performance and that the use of linguistic evidence, multi-agent collaboration, and fine-tuning can improve prediction distributions across proficiency levels. However, in practical AWE applications, scoring accuracy alone is not sufficient. Since feedback is a core output of LLM-based AWE, it is also necessary to examine whether the system can generate reliable feedback when its score predictions are correct and whether the feedback remains useful when its score predictions are incorrect. Therefore, we conduct an additional experiment to examine the feedback quality of the LLM-based AWE system when the predicted scores are correct and when they are incorrect. Cases where the model correctly predicts the level are treated as positive samples, whereas misclassified cases are treated as negative samples. Negative samples are further divided into lenient misclassifications (the model assigns a higher level than the human reference) and severe misclassifications (the model assigns a lower level than the human reference). We then evaluate each model's performance on both positive and negative samples using the three dimensions of coverage of objective evidence, internal consistency, and overall feedback quality. The results are presented in Table 7.

Across all five settings, models generate higher-quality feedback when score predictions are correct. For example, in Setting 2, Qwen3-80B achieves an objective evidence coverage score of 4.9779, an internal consistency score of 4.9834, and a overall feedback quality score of 4.9820 on positive samples, all of which exceed its corresponding scores on negative samples (4.8718, 4.8866, and 4.9076, respectively). This finding demonstrates that correct score predictions are generally accompanied by higher-quality feedback and vice versa. Within the negative samples, our results show that after introducing advanced prompting, multi agent collaboration, or fine tuning, the severe samples tend to score higher than the lenient samples on all three feedback metrics. Based on case studies of a set of lenient and severe examples, we observed that lenient samples often elicit a more praise oriented generation style. As a result, the model identifies fewer linguistic problems and cites fewer textual evidence points. By contrast, in severe samples, the model adopts a more critical diagnostic stance, provides more text grounded evidence, and is therefore more likely to receive higher ratings. Sawaki et al. (2025) report a similar pattern in that harsher LLM scoring is associated with more item by item checking of the criteria, closer attention to textual details, and heavier reliance on whether specific points are explicitly stated. Therefore, the higher feedback scores

Table 7

Feedback performance of the Chinese L2 LLM-based AWE system on positive and negative samples.

Models	Positive Samples			Negative Samples								
	Objective Evidence Coverage	Internal Consistency	Overall Feedback Quality	Objective Evidence Coverage			Internal Consistency			Overall Feedback Quality		
				Lenient Samples	Severe Samples	Overall Samples	Lenient Samples	Severe Samples	Overall Samples	Lenient Samples	Severe Samples	Overall Samples
Setting 1: General-purpose LLMs using the general prompt												
DeepSeek-Chat-V3	2.9559	4.1092	3.8946	2.913	2.7756	2.7778	3.9217	3.8416	3.7733	3.858	3.7657	3.7067
GPT-4.1-mini	3.0545	4.2236	3.6473	3.0236	2.8586	2.9231	4.0354	3.9091	3.9585	3.7992	3.7197	3.7508
Qwen3-80B	4.4107	4.8886	4.7679	4.2079	4.3878	4.3211	4.7475	4.9417	4.8697	4.6832	4.7347	4.7156
Setting 2: General-purpose LLMs using the advanced prompt												
DeepSeek-Chat-V3	4.8165	4.8228	4.7041	4.7958	4.689	4.552	4.8208	4.6622	4.5485	4.7208	4.6321	4.4903
GPT-4.1-mini	4.8979	4.9085	4.5305	4.809	4.8531	4.7537	4.8315	4.8475	4.7574	4.7584	4.6243	4.5882
Qwen3-80B	4.9779	4.9834	4.982	4.7628	4.8955	4.8718	4.7905	4.8955	4.8866	4.83	4.8955	4.9076
Setting 3: Multi-agent systems using the advanced prompt + meta prompt												
GPT-4.1-mini + Qwen3-80B	4.7204	4.8108	4.6606	4.3362	4.6933	4.4271	4.3799	4.7269	4.4646	4.4192	4.6681	4.4542
-> DeepSeek-Chat-V3 (meta agent)												
DeepSeek-Chat-V3 + Qwen3-80B -> GPT-4.1-mini (meta agent)	4.9375	4.9637	4.9041	4.8511	4.9461	4.8281	4.874	4.9336	4.834	4.8817	4.9336	4.8379
DeepSeek-Chat-V3 + GPT-4.1-mini -> Qwen3-80B (meta agent)	4.9874	4.9944	4.9874	4.9151	4.9622	4.963	4.9292	4.9653	4.9691	4.9387	4.9688	4.9733
Setting 4: Fine-tuned LLMs using the advanced prompt												
Fine-tuned GPT-4.1-nano	4.8804	4.876	4.8793	4.5782	4.9728	4.7374	4.5646	4.966	4.7273	4.6327	4.9592	4.7576
Fine-tuned Qwen3-8B	4.9874	4.9895	4.9854	4.8684	4.9921	4.8689	4.8596	4.9841	4.8607	4.9035	4.9921	4.8852
Setting 5: Fine-tuned multi-agent system using the advanced prompt + meta prompt												
Fine-tuned GPT-4.1-nano + Fine-tuned Qwen3-8B -> Qwen3-80B (meta agent)	4.9846	4.9867	4.9826	4.936364	4.9937	4.9689	4.927273	4.992	4.9644	4.945455	4.9951	4.9733

observed in severe cases likely reflect stronger evidence grounding and more comprehensive issue identification.

However, the generated feedback still has instructive values even when the model's score prediction is incorrect. We conduct a case study to illustrate the feedback produced by the fine-tuned Qwen3-8B on a negative sample²⁰ (see Appendix I for details). In this case, the essay's true level is B, but the fine-tuned Qwen3-8B predicts it as A. Two linguistics PhD students manually validated the feedback without access to the original score level. They both agreed that the model's reasoning and feedback were well-founded and acceptable. They also noted that the feedback still provided objective insights into the essay, which can help L2 Chinese learners improve their writing quality and assist human teachers in quickly identifying the weaknesses of the essay.

Overall, our LAWE-CL2 system (setting 5) achieves a QWK of 0.7989 and an F1-score of 0.8149 (see Table 3, setting 5), and this high level of prediction accuracy ensures the quality of its feedback. Moreover, even on negative samples, the system's coverage of objective evidence, internal consistency, and overall feedback quality all remain above 4.9. These results indicate that the feedback produced by the system possesses robust pedagogical value and holds promise for fostering high user confidence in its feedback in future applications.

5. Discussion

The proposed LAWE-CL2 has been validated across diverse experimental settings with mainstream baseline LLMs, underscoring its broad applicability. Moreover, empirical evidence from Chinese L2 writing confirms the potential of GenAI in language assessment, further reinforcing the generalizability of LLMs in evaluating writing.

5.1. Discussion of RQ1: Construct-grounded prompting

The findings for RQ1 show that embedding detailed multidimensional linguistic information into the prompts substantially improves LLM performance in both scoring and feedback generation for Chinese L2 writing. This finding not only responds to calls from Mizumoto and Eguchi (2023), Liu et al. (2025b), and Li and Liu (2024) for greater integration of linguistic features in LLM-based AWE, but also further confirms that linguistic features remain necessary in automated writing evaluation (Li & Ng, 2024). In the present study, these improvements are reflected not only in scoring metrics such as QWK, EAR, and F1-score, but also in higher levels of objective evidence coverage, internal consistency, and overall feedback quality. This suggests that linguistic features improve not only scoring accuracy but also the evidential grounding of model evaluations.

We argue that the main reason for this improvement lies in the way multidimensional linguistic indices are used in the prompts. In this study, these linguistic features are operationalized as interpretable and construct-relevant scoring guidance and explicitly incorporate into the model's evaluative reasoning. Specifically, the advanced prompt explains the linguistic meaning of each index, its empirically observed relationship with proficiency differences in Chinese L2 writing, and its relation to benchmark distributions derived from essays at different proficiency levels. In other words, the model is given not isolated numerical values but a reference framework directly related to judgments of writing ability. This framework enables the model to establish more explicit links between linguistic features in the essay and the criteria associated with different score levels. Such a design likely makes the model's judgments less impressionistic and more closely tied to specific linguistic evidence, thereby improving scoring accuracy and stability.

This prompt design also helps explain why feedback quality improves at the same time. Because both scoring and feedback are grounded in the same set of linguistic evidence, the model is no longer limited to broad, impressionistic comments when generating feedback. Instead, it is able to provide diagnostic feedback on specific linguistic dimensions. For example, the prompts explicitly specify how MDD reflects the development of syntactic complexity in Chinese L2 writing, how topic-chain density indicates stronger topic-prominent discourse organization, and how patterns of connective use relate to the development of paratactic coherence in Chinese. Such prompts allow the model to translate abstract scoring criteria into observable textual features and to generate feedback that is more explanatory and diagnostic. In this sense, linguistic features are not merely supplementary to the prompt. Instead, they provide an explicit evidential basis for both scoring and feedback.

One important contribution of the present study lies in embedding linguistic indices and their benchmark information directly into the prompts, rather than treating them as post hoc correlates of essay scores, allowing the model to better understand their construct meaning during scoring and capture the latent associations between scores and linguistic features (Atkinson & Palma, 2025). As a result, the model can not only attend to these linguistic features but also better understand why they are associated with particular score levels. This makes it easier to align the essay's linguistic performance with the scoring criteria. If only general scoring criteria are provided, the model may not adequately attend to Chinese typological features such as topic chains, paratactic coherence, and run-on sentence patterns. By explicitly incorporating indices with clear linguistic-construct relevance into the prompts, the model's judgments are more likely to reflect the actual characteristics of Chinese L2 writing.

Our qualitative analysis of the generated feedback also supports this interpretation. As shown in Fig. 3, taking the syntactic dimension as an example, the model is able to reason about the original essay by drawing on the Chinese typological features specified in the prompt and to explicitly identify run-on sentence patterns and topic-prominent structures in the text. This suggests that the model does not use linguistic terms mechanically, but instead integrates linguistic knowledge, scoring criteria, and textual evidence into a relatively coherent evaluative judgment under the guidance of the advanced prompt.

²⁰ As the feedback texts generated by the model are relatively lengthy and the main text is limited in space, the detailed feedback is provided in Appendix I.

Key:
 Taking the feedback on an essay titled "A Job Application Letter" as an example
 Feedback incorporating Chinese typological features:

4. 评审意见-句法:
 该作文句法呈现高复杂度但低多样性的矛盾特征。MLS (18.0667) 显著低于 A 档均值 (32.991), 表明句子整体偏短, 缺乏长句的复杂结构; NCPS (1.6) 远低于 A 档均值 (3.051), 说明作者极少使用流水句或多分句结构, 句式结构单一, 未能充分体现汉语复合特征。MLC (11.2917) 接近 A 档均值 (11.116), 但因 NCPS 低, 句子虽短却缺乏层次, 语义承载集中, 显得生硬。句法多样性方面, TOTAL_RTTR (8.1276) 和 UNIQUE_RTTR (2.6056) 均高于 A 档均值 (8.513 和 2.542), 说明搭配多样性良好, 尤其在汉语特有搭配 (CN_RTTR=2.0, PC_RTTR=2.2361) 上表现突出, 如“吸烟”等量名使用准确, 但 VO_RATIO (0.1857) 和 SP_RATIO (0.2286) 低于 A 档均值 (0.209 和 0.183), 说明主谓宾结构使用比例偏低, 符合一定的汉语话题突出特征, 但因 NCPS 低, 未能形成有效话题链。MDD (2.1006) 低于 A 档均值 (2.969), 表明句法结构依赖简单线性排列, 缺乏深层依存关系。话题链 句数比例 (0.4) 高于 A 档均值 (0.241), TCMD (1.8984) 和 MLTTCU (12.6667) 也高于 A 档均值, 说明作者在局部段落中已能构建完整的话题链, 如“吸烟者→身体免疫力减退→眼红、手颤抖→没精神→影响生活→工作效率”, 这是本文最突出的句法优势, 建议在保持话题链优势的基础上, 增加 NCPS, 尝试将多个短句通过“也”“还”“而且”等连接词串联为流水句, 如“吸烟者不仅身体免疫力下降, 还常出现眼红、手颤抖、没精神等状况, 这些都会影响他们的生活质量和工作效率”, 以提升句法复杂度与自然度。

Note: Highlighted segments illustrate how the model grounds its feedback in Chinese typological features (e.g., run on sentence style, parataxis, and subject omission). The full feedback and an English translation are provided in Appendix I.

Fig. 3. LLM Generated Feedback Highlighting Chinese Typological Features.

5.2. Discussion of RQ2: Multi-agent adjudication

The findings for RQ2 show that all three multi-agent configurations outperform their corresponding single-agent systems, regardless of which LLM serves as the meta agent. This suggests that the gains in score prediction for Chinese L2 writing cannot be attributed solely to the capability of any single LLM, but are also closely related to the collective adjudication mechanism enabled by multi-agent collaboration. The underlying logic is analogous to the multi-rater scoring and adjudication procedures used in operational writing assessment, which are intended to reduce rater-specific bias and improve rating stability. This finding is consistent with Cheng and Li (2025) and further suggests that multi-agent collaboration represents a promising direction for the future development of multilingual automated writing evaluation systems.

The advantage of multi-agent systems over single-agent systems can also be explained by their adjudication logic. In a single-agent setting, the final score depends on a single model's judgment, which may make the result somewhat more vulnerable to model-specific scoring biases or occasional misjudgments. In contrast, in our framework, the meta agent makes the final decision after considering the scores and feedback generated by the two primary agents. This allows the final decision to draw on multiple evaluative perspectives rather than relying solely on a single model's judgment. As a result, the multi-agent framework reduces dependence on any one model's scoring tendency and contributes to greater scoring stability.

The level-specific results further support this account. Compared with single-agent systems, multi-agent systems not only improve overall QWK, EAR, and F1-score, but also show more balanced prediction patterns across proficiency levels. General-purpose LLMs may tend to produce safer and more moderate outputs, partly because many of them have undergone reinforcement learning from human feedback. In writing assessment, this tendency may make them less willing to assign more extreme high- or low-proficiency judgments, leading predictions to cluster around the middle level. Under the multi-agent setting, however, the prediction proportions for low- and high-level essays increase, while the proportion of Level B predictions decreases. This suggests that, when multiple evaluative perspectives are available, the system becomes less dependent on a middle-level default and is better able to identify essays at both ends of the proficiency continuum.

The internal-consistency analysis also supports this interpretation. The meta agent shows high consistency with both primary agents, and the small difference between the two pairwise consistency values suggests that it does not strongly favor either side. This indicates that the multi-agent framework does not simply replicate the judgment of a single model, but instead forms a more balanced and integrated decision on the basis of multiple evaluations. The findings for RQ2 therefore suggest that the advantage of multi-agent systems lies not only in stronger overall performance, but also in their closer alignment with the multi-rater logic of authentic writing assessment. By introducing comparative reasoning and adjudication, the framework can reduce the influence of single-model bias and provide a more reliable and balanced basis for Chinese L2 writing assessment across proficiency levels.

5.3. Discussion of RQ3: Fine-tuning and domain adaptation

RQ3 examines whether fine-tuning yields greater gains than prompting alone in Chinese L2 automated writing evaluation. Although Mizumoto and Eguchi (2023) showed that GPT could broadly follow human-like scoring patterns in TOEFL essay scoring and distinguish essays of different quality levels to some extent, its exact agreement with human ratings remained limited. This may be partly because their study relied only on a zero-shot setting, which may have affected the stability of the results. By comparison, our findings are consistent with those of Wang and Gayed (2024), Liu et al. (2025a), and Liu et al. (2025b), and further confirm that, in LLM-based AWE, fine-tuning offers more substantial advantages than relying on prompting alone. Our results show that fine-tuned models clearly outperform models that rely only on the advanced prompt, both in scoring performance and in feedback quality. In

other words, although prompt engineering is important, it is usually not sufficient on its own for writing assessment, a task that involves explicit criteria, strong demands for evidence, and considerable complexity. Task-specific training remains a crucial way to improve performance.

A particularly notable finding is that fine-tuning enables two relatively small LLMs (GPT-4.1-nano and Qwen3-8B) to outperform their original larger counterparts. We further observe a consistent pattern in which stronger baseline model performance is associated with greater AWE potential, whether the models are used as meta agents or fine-tuned. These results suggest that final performance in AWE cannot be explained by model size alone. What matters more is whether the model has learned the judgment logic required for the task through fine-tuning. Although general-purpose large models have stronger language understanding and generation abilities, these strengths do not necessarily translate into higher scoring agreement and better feedback quality without training aligned specifically with Chinese L2 writing assessment. By contrast, once a smaller model has secured the scoring criteria, feedback patterns, and the links between linguistic evidence and score-level judgments through fine-tuning, it may perform more stably and accurately on this task than a larger model that relies only on prompting.

This further suggests that the role of fine-tuning is not simply to help the model adapt to the formal requirements of the task. More importantly, it allows the model to learn knowledge structures, scoring rationales, and feedback patterns that are more closely aligned with Chinese L2 writing assessment from domain-specific data. For this reason, a smaller fine-tuned model may outperform a larger model that has not been fine-tuned (Liu et al., 2025b). The synthetic feedback data used in the present study also play an important role here. During fine-tuning, the model learns not isolated score labels, but the full process by which linguistic performance in an essay corresponds to a particular score-level judgment and how appropriate feedback can be generated on that basis. As a result, the fine-tuned models improve not only in QWK, EAR, and F1-score, but also in objective evidence coverage, internal consistency, and overall feedback quality. This suggests that scoring and feedback are not treated as two separate tasks in the present study. Rather, they are modeled as two outputs grounded in the same set of evaluative criteria. Because the model has learned both how to judge and how to explain, it is able to achieve stronger overall performance in the AWE task.

5.4. General Discussion

The findings of the present study offer broader implications for AI-driven automated writing evaluation systems. First, the effectiveness of LLM-based AWE cannot depend solely on a model's general generative ability, but must be grounded in a clearer evaluative basis. Both the substantial gains produced by embedding linguistic features into prompts in RQ1 and the further improvements achieved through fine-tuning in RQ3 suggest that a high-quality LLM-based AWE system should not focus only on producing fluent outputs or plausible responses. It must also develop stable grounds for judging writing ability. This is consistent with earlier work in writing assessment, which has emphasized that scoring should be based on clearly defined constructs and interpretable evidence (Bennett & Bejar, 1998; Kane, 2006; Correnti et al., 2022). It also suggests that future research on LLM-based AWE should move beyond the question of how to make a model assign scores and focus more on how to enable LLMs to evaluate writing on the basis of reasonable and interpretable criteria. In other words, the development of LLM-based AWE should not remain limited to comparing model size, prompting strategies, or isolated scoring metrics. Instead, it should return to the long-standing core concerns of automated writing evaluation by asking whether scoring is grounded in valid linguistic evidence (Hao et al., 2024), whether feedback reflects actual textual evidence, and whether system outputs can support pedagogical interpretation and learning improvement.

Second, the present study supports understanding AWE as an integrated system of scoring and feedback rather than as two separate modules. Our analysis of positive and negative samples shows that feedback is usually of higher quality when score predictions are correct. Even in misclassified cases, however, the system's feedback still retains a considerable degree of objectivity and pedagogical value. This suggests that, within an LLM framework, scoring and feedback should not be understood simply as a primary task and a subsidiary output. Rather, they are better understood as two related outputs grounded in the same evaluative basis. This means that AWE system design should not focus only on scoring accuracy, but should also consider whether feedback genuinely reflects textual evidence and whether it is diagnostically meaningful and pedagogically usable (Wilson & Roscoe, 2020). In instructional settings, the educational value of feedback is often as important as scoring accuracy, and in some low-stakes applications it may even be more important.

Overall, the general discussion of the present study suggests that the future development of LLM-based AWE systems should not be framed only in terms of linear gains in model performance. Equal attention should be paid to whether evaluative criteria are well defined, whether feedback has instructional value, whether learners at different proficiency levels are treated fairly, and whether the system is appropriate for real teaching and assessment contexts.

6. Conclusion

This study examines a multi-agent LLM-based AWE system with linguistic features and fine-tuning for Chinese L2 writing. The results show its effectiveness and provide empirical evidence supporting the generalizability of LLMs in writing assessment. To reveal the effectiveness of LAWE-CL2, a Chinese L2 LLM-based AWE system that integrates multidimensional linguistic features, a multi-agent scoring framework, and fine-tuning, we conduct five comparative experiments. The results reveal that embedding multidimensional linguistic features into prompts substantially improves AWE performance, underscoring the importance of linguistic features in L2 AWE and pointing to the need for further exploration of indices that more effectively capture L2 writing development. The multi-agent scoring framework is shown to substantially enhance Chinese L2 AWE performance and ensure the fairness and consistency of the scoring process. Moreover, fine-tuning with synthetic feedback data further strengthens the performance of smaller-scale LLMs,

enabling them to surpass their baseline versions. In addition, this study integrates linguistic features, prompt engineering, and LLM fine-tuning into a unified pipeline, offering a reproducible methodological approach. We argue that this approach provides new empirical evidence for the feasibility of LLM-based AWE in Chinese L2 writing. It also has practical value in reducing the workload of teachers, improving educational efficiency, and offering learners personalized, intelligent feedback and guidance.

Nevertheless, several limitations of the present study should be acknowledged, which also point to clear directions for future research. First, the analyzed essays were written by L2 speakers who took the old version of the HSK between 1992 and 2005. Given the long time span of more than two decades and the subsequent updates to the HSK, future work should validate the proposed system on more recent HSK writing corpora and other contemporary Chinese L2 learner writing datasets to further test the generalizability of our conclusions. Second, our scoring task was formulated as a three-level proficiency classification (low, medium, high) rather than a finer-grained continuous score scale. While the present findings provide new evidence for the potential of LLM-based assessment in Chinese L2 writing, we caution against immediate deployment in large-scale, high-stakes contexts. Future studies should move beyond level prediction and generate numerical scores that better match operational needs and allow a more stringent evaluation of reliability, validity, and fairness. In addition, further progress will require more comprehensive evaluation and modeling of feedback. This includes systematic comparisons between human-generated and AI-generated feedback, the construction of a gold-standard Chinese L2 feedback dataset to support more rigorous benchmarking and model improvement, and deeper incorporation of multidimensional linguistic features to strengthen construct alignment in both scoring and diagnostic feedback. Finally, future work should examine robustness and fairness across learner populations and task conditions by using corpora with more balanced genre distributions and by testing whether performance varies by genre and learners' native language backgrounds.

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CRedit authorship contribution statement

Xuelin Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qihao Yang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuxin Hao:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Zhijun Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration. **Sijia Guo:** Writing – review & editing, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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During the preparation of this work, we used ChatGPT (OpenAI) to improve the English academic language of the manuscript. After using this tool, we thoroughly reviewed and revised the text as necessary, and we take full responsibility for all aspects of the published content.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.asw.2026.101051](https://doi.org/10.1016/j.asw.2026.101051).

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